

Sesión 8

Privacidad, seguridad, y administración

Agenda

- Mejores prácticas de manejo de tu cuenta
- Autenticación a 2 pasos
- Gestión de permisos
- Visibilidad de repositorios
- Opciones de la pestaña de seguridad
- Habilitar/Deshabilitar features en tu cuenta de GitHub

Mejores prácticas de manejo de
tu cuenta

Autenticación en GitHub

- Puedes acceder de manera segura a los recursos de tu cuenta si te autentificas en GitHub, utilizando diferentes credenciales en función del método usado para la autenticación.
- Formas de acceder a los recursos en GitHub:
 - Mediante el navegador
 - A través de GitHub Desktop
 - Mediante otra aplicación de escritorio (VSCode por ejemplo)
 - Por medio del API
 - Usando la línea de comandos (CLI)
- Formas de autenticación:
 - Usuario y contraseña
 - Token de acceso personal (PAT)
 - Passkey
 - Autenticación de 2 factores
 - SAML SSO
 - Llave SSH

Usuario y contraseña

- Se recomienda utilizar un gestor de contraseñas para generar una contraseña aleatoria.
- GitHub puede solicitar verificación adicional al acceder a recursos desde un dispositivo no reconocido (computadora, teléfono, navegador, perfil de navegador).
 - Solo si la autenticación de 2 factores no está habilitada.
 - Solicitud de verificación a GitHub Mobile.
 - Código de verificación (válido por una hora) a tu correo principal y de respaldo si no tienes instalado GitHub Mobile.
- Si recibes un código de verificación de GitHub que tú no solicitaste, considera tu contraseña comprometida y cámbiala de inmediato. Además, toma los pasos necesarios para asegurar tu cuenta.

https://github.com/password_reset

Recomendaciones y requisitos de contraseña

Requisitos

- Mínimo 8 caracteres si incluye números y letras minúsculas
- Mínimo 15 caracteres si incluye cualquier combinación de caracteres

Recomendaciones

- Usa un gestor de contraseñas para contraseñas de 15 caracteres mínimo
- Usa un password único para GitHub
- Configura la autenticación de 2 factores para tu cuenta personal
- Opcionalmente, añade un passkey a tu cuenta para un inicio de sesión seguro, sin contraseña.
- No compartas tu contraseña, incluso ni a un colaborador potencial.

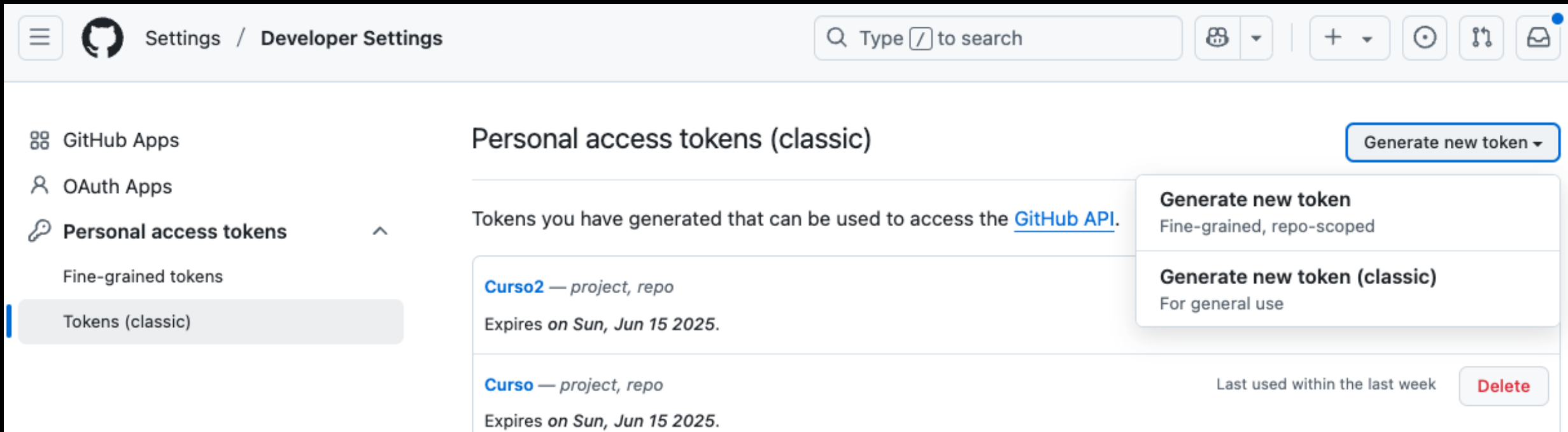
Verificación de contraseña

- GitHub verifica si tu contraseña es considerada débil de acuerdo a datasets como HaveIBeenPwned.
- GitHub verifica tu contraseña cuando la escribes, pero no la almacena en texto plano.
- Solo puedes usar tu contraseña para iniciar sesión en el navegador.
- La autenticación por otros medios (API, CLI) requiere el uso de otras credenciales.
- Si Git solicita tu contraseña, utiliza tu PAT o utiliza Git Credential Manager.

Tokens de acceso personal (PAT)

- Utilizado para autenticarse en GitHub en la CLI o con la API
- Propósito: acceder a recursos de GitHub en tu nombre.
 - Para acceder a recursos a nombre de una organización o integraciones de larga duración, utiliza una GitHub App
- Un token está limitado por los ámbitos o permisos concedidos.
- Tipos:
 - Fine-grained PAT (recomendados)
 - PAT clásico
- Considera un token como una contraseña.
- Métodos más seguros:
 - CLI: Autenticación desde GitHub CLI, Git Credential Manager
 - GitHub Actions: valor GITHUB_TOKEN integrado en los workflows (solo tendrá acceso a recursos dentro del repositorio que contiene el workflow)

¿Dónde crear un PAT?



The screenshot shows the GitHub Developer Settings page. The left sidebar contains a menu with 'GitHub Apps', 'OAuth Apps', and 'Personal access tokens'. Under 'Personal access tokens', there are two sub-items: 'Fine-grained tokens' and 'Tokens (classic)', with the latter being selected. The main content area is titled 'Personal access tokens (classic)' and includes a 'Generate new token' button. Below this, a text block states: 'Tokens you have generated that can be used to access the [GitHub API](#).' A dropdown menu is open, showing two options: 'Generate new token' (Fine-grained, repo-scoped) and 'Generate new token (classic)' (For general use). Below the dropdown, a list of existing tokens is shown. The first token is named 'Curso2' with scope 'project, repo' and expires on 'Sun, Jun 15 2025'. The second token is named 'Curso' with scope 'project, repo' and also expires on 'Sun, Jun 15 2025'. To the right of the second token, it says 'Last used within the last week' and there is a 'Delete' button.

Settings / Developer Settings

Search Type / to search

GitHub Apps

OAuth Apps

Personal access tokens

Fine-grained tokens

Tokens (classic)

Personal access tokens (classic)

Generate new token

Tokens you have generated that can be used to access the [GitHub API](#).

- Generate new token
Fine-grained, repo-scoped
- Generate new token (classic)
For general use

Curso2 — project, repo	
Expires on Sun, Jun 15 2025.	
Curso — project, repo	Last used within the last week
Expires on Sun, Jun 15 2025.	Delete

Fine-grained PATs

- Tokens avanzados de seguridad
- Se puede limitar los repositorios a los que puede acceder
- A cada token se le conceden permisos específicos y detallados, que ofrecen más control que los ámbitos concedidos a PATs clásicos.
- Limitaciones: No se pueden usar Fine-grained PATs para
 - Contribuir en repositorios públicos donde el usuario no es un miembro
 - Contribuir en repositorios donde el usuario es un colaborador externo
 - Acceder a múltiples organizaciones a la vez
 - Acceder a GitHub Packages
 - Llamar la API Checks
 - Acceder a Proyectos cuyo dueño es una cuenta de usuario

New fine-grained personal access token

Create a fine-grained, repository-scoped token suitable for personal API use and for using Git over HTTPS.

Token name *

Demo GitHub Models

✔ 'Demo GitHub Models' is available.

A unique name for this token. May be visible to resource owners or users with possession of the token.

Description

Token para demostrar GitHub Models

Resource owner

 icebeam7 ▾

The token will only be able to make changes to resources owned by the selected resource owner. Tokens can always read all public repositories.

Expiration

 30 days (Jun 19, 2025) ▾

The token will expire on the selected date

Repository access

☐ Public repositories

Read-only access to public repositories.

☐ All repositories

This applies to all current and future repositories you own. Also includes public repositories (read-only).

☒ Only select repositories

Select at least one repository. Max 50 repositories. Also includes public repositories (read-only).

 Select repositories 2 ▾

Permissions

Choose the minimal permissions necessary for your needs. [Learn more about permissions.](#)

Repository permissions

Repository permissions permit access to repositories and related resources.

⌵

Account permissions 1 selected

User permissions permit access to resources under your personal GitHub account.

⌵

Generate token

Cancel

This token will be ready for use immediately.

Repository permissions

Repository permissions permit access to repositories and related resources.



Actions

Workflows, workflow runs and artifacts. [Learn more.](#)

Access: No access ▾

Administration

Repository creation, deletion, settings, teams, and collaborators. [Learn more.](#)

- ✓ No access
- Read-only
- Read and write

Attestations

Create and retrieve attestations for a repository. [Learn more.](#)

Access: No access ▾

Code scanning alerts

View and manage code scanning alerts. [Learn more.](#)

Access: No access ▾

Codespaces

Create, edit, delete and list Codespaces. [Learn more.](#)

Access: No access ▾

Codespaces lifecycle admin

Manage the lifecycle of Codespaces, including starting and stopping. [Learn more.](#)

Access: No access ▾

Codespaces metadata

Access Codespaces metadata including the devcontainers and machine type. [Learn more.](#)

Access: No access ▾

Codespaces secrets

Restrict Codespaces user secrets modifications to specific repositories. [Learn more.](#)

Access: No access ▾

Account permissions

User permissions permit access to resources under your personal GitHub account.



Block another user

View and manage users blocked by the user. [Learn more.](#)

Access: No access ▼

Codespaces user secrets

Manage Codespaces user secrets. [Learn more.](#)

Access: No access ▼

Copilot Chat

This application will receive your GitHub ID, your GitHub Copilot Chat session messages (not including messages sent to another application), and timestamps of provided GitHub Copilot Chat session messages. This permission must be enabled for Copilot Extensions. [Learn more.](#)

Access: No access ▼

✓ No access

Read-only

Copilot Editor Context

This application will receive bits of Editor Context (e.g. currently opened file) whenever you send it a message through Copilot Chat. [Learn more.](#)

Access: No access ▼

Email addresses

Manage a user's email addresses. [Learn more.](#)

Access: No access ▼

Events

View events triggered by a user's activity. [Learn more.](#)

Access: No access ▼

Followers

A user's followers [Learn more.](#)

Access: No access ▼

GPG keys

View and manage a user's GPG keys. [Learn more.](#)

Access: No access ▼

Gists

Create and modify a user's gists and comments. [Learn more.](#)

Access: No access ▼

Demo 1 – Fine-grained PATs para consumir GitHub Models

New personal access token

×

Your new personal access token **Demo GitHub Models** will be ready for use immediately. It will expire on **Thursday, June 19, 2025**.

Demo GitHub Models grants you **1 account permission**:

Models

Read-only

Cancel

Generate token

<https://docs.github.com/en/github-models/use-github-models/prototyping-with-ai-models>

Fine-grained personal access tokens

Generate new token

These are fine-grained, repository-scoped tokens suitable for personal [API](#) use and for using Git over HTTPS.



Demo GitHub Models

Never used • Expires on Thu, Jun 19 2025

Delete



Make sure to copy your personal access token now as you will not be able to see this again.

github_pat_11AB0CFXY



AINetLatamToken

Never used • ⚠️ [This token has expired](#)

Delete



ReactorDemo

Never used • ⚠️ [This token has expired](#)

Delete

```
4 import os
5 from openai import OpenAI
6
7 token = os.environ["GITHUB_TOKEN"]
8 endpoint = "https://models.github.ai/inference"
```

⚙️ .env

```
1 GITHUB_TOKEN="github_pat_11ABOCFXy0pxPaLQzYxwvb_cD
2
```

```
● @icebeam7 → /workspaces/codespaces-models (main) $ /usr/local/python/current/bin/python /workspaces/
codespaces-models/samples/python/openai/basic.py
The capital of France is Paris.
○ @icebeam7 → /workspaces/codespaces-models (main) $
```


PATs clásico

- Algunas características solo funcionan con tokens PAT (clásico).
 - Escritura en repositorios públicos de diferente dueño / organización (aún sin ser miembro).
 - Colaboradores externos solo pueden usar PAT (clásico) en repositorios de la organización en la que colaboran.
 - Algunos endpoints del API REST
- Un PAT clásico es menos seguro.
 - Se otorga acceso a todos los repos (de tu cuenta y de las organizaciones a las que tienes acceso)
 - GitHub elimina automáticamente PATs (clásico) que no se han usado en un año. Se recomienda agregar caducidad a los tokens.

New personal access token (classic)

Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to [authenticate to the API over Basic Authentication](#).

Note

PAT para workflow y discusiones automáticas

What's this token for?

Expiration

 30 days (Jun 19, 2025) ▾

The token will expire on the selected date

Select scopes

Scopes define the access for personal tokens. [Read more about OAuth scopes](#).

☐ repo	Full control of private repositories
☐ repo:status	Access commit status
☐ repo_deployment	Access deployment status
☒ public_repo	Access public repositories
☐ repo:invite	Access repository invitations
☐ security_events	Read and write security events
☐ workflow	Update GitHub Action workflows
☐ write:packages	Upload packages to GitHub Package Registry
☐ read:packages	Download packages from GitHub Package Registry
☐ delete:packages	Delete packages from GitHub Package Registry

☒ write:discussion	Read and write team discussions
☒ read:discussion	Read team discussions
☐ admin:enterprise	Full control of enterprises
☐ manage_runners:enterprise	Manage enterprise runners and runner groups
☐ manage_billing:enterprise	Read and write enterprise billing data
☐ read:enterprise	Read enterprise profile data
☐ scim:enterprise	Provisioning of users and groups via SCIM
☐ audit_log	Full control of audit log
☐ read:audit_log	Read access of audit log
☐ codespace	Full control of codespaces
☐ codespace:secrets	Ability to create, read, update, and delete codespace secrets
☐ copilot	Full control of GitHub Copilot settings and seat assignments
☐ manage_billing:copilot	View and edit Copilot Business seat assignments
☐ write:network_configurations	Write org hosted compute network configurations
☐ read:network_configurations	Read org hosted compute network configurations
☐ project	Full control of projects
☐ read:project	Read access of projects
☐ admin:gpg_key	Full control of public user GPG keys
☐ write:gpg_key	Write public user GPG keys
☐ read:gpg_key	Read public user GPG keys
☐ admin:ssh_signing_key	Full control of public user SSH signing keys
☐ write:ssh_signing_key	Write public user SSH signing keys
☐ read:ssh_signing_key	Read public user SSH signing keys

Demo 2 – PAT clásico para un workflow de GitHub Actions que genera una discusión

Personal access tokens (classic)

Generate new token ▾

Tokens you have generated that can be used to access the [GitHub API](#).

 Make sure to copy your personal access token now. You won't be able to see it again!

✓ ghp_ny4MS0pit[REDACTED]		Delete
Curso2 — <i>project, repo</i>	Never used	Delete
Expires on Sun, Jun 15 2025.		

GraphQL API

About the GraphQL API

Public schema

Breaking changes

Changelog

Explorer

Rate and node limits

Reference

Guides

GitHub GraphQL API

Signed in as **icebeam7**. You're ready to explore!

[Sign out](#)

Heads up! GitHub's GraphQL Explorer makes use of your *real, live, production* data.


```

1 query
2 {
3   repository(owner: "icebeam7", name: "demos-sesion-8") {
4     id # RepositoryID
5     name
6     discussionCategories(first: 10) {
7       nodes {
8         id # CategoryID
9         name
10      }
11    }
12  }
13 }

```



Variables

Headers

1



- + GraphQL

```

{
  "data": {
    "repository": {
      "id": "R_kgD00tXsoA",
      "name": "Demos-Sesion-8",
      "discussionCategories": {
        "nodes": [
          { ↔ },
          { ↔ },
          { ↔ },
          { ↔ },
          { ↔ },
          {
            "id": "DIC_kwD00tXsoM4CqYil",
            "name": "Show and tell"
          },
          {
            "id": "DIC_kwD00tXsoM4CqYit",
            "name": "Últimos commits"
          }
        ]
      }
    }
  }
}

```

Actions secrets and variables

Secrets and variables allow you to manage reusable configuration data. Secrets are **encrypted** and are used for sensitive data. [Learn more about encrypted secrets](#). Variables are shown as plain text and are used for **non-sensitive** data. [Learn more about variables](#).

Anyone with collaborator access to this repository can use these secrets and variables for actions. They are not passed to workflows that are triggered by a pull request from a fork.

Secrets

Variables

Environment secrets

This environment has no secrets.

Manage environment secrets

Repository variables

New repository variable

Name	Value	Last updated
SUMMARY_FILE	RESUMEN.MD	12 minutes ago

Repository secrets

New repository secret

Name	Last updated
DISCUSSION_CATEGORY_ID	1 hour ago
MY_PERSONAL_ACCESS_TOKEN	1 hour ago

```

1  name: Crear discusión con resumen de commits
2
3  on:
4    workflow_dispatch:
5
6  jobs:
7    crear-discusion:
8      runs-on: ubuntu-latest
9      env:
10        SUMMARY_FILE : ${vars.SUMMARY_FILE}
11
12    steps:
13      - name: Checkout del repositorio
14        uses: actions/checkout@v4
15        with:
16          fetch-depth: 0
17          fetch-tags: true
18
19      - name: Obtener últimos 5 commits y guardar como Markdown
20        id: get_commits
21        run: |
22          echo "# Últimos 5 commits de la rama main" > $SUMMARY_FILE
23          echo "" >> $SUMMARY_FILE
24          git fetch origin main
25          git log origin/main -n 5 --pretty=format:"- %h %s (%an, %ad)" --date=short >> $SUMMARY_FILE
26          echo $SUMMARY_FILE

```

```

28      - name: Obtener repositoryId con GraphQL
29        id: get_repo_id
30        env:
31          GH_TOKEN: ${secrets.MY_PERSONAL_ACCESS_TOKEN}
32        run: |
33          REPO_OWNER=$(echo "${github.repository}" | cut -d '/' -f1)
34          REPO_NAME=$(echo "${github.repository}" | cut -d '/' -f2)
35
36          QUERY=$(cat <<EOF
37          {
38            "query": "query { repository(owner: \"$REPO_OWNER\", name: \"$REPO_NAME\") { id } }"
39          }
40          EOF
41          )
42
43          REPO_ID=$(curl -s -H "Authorization: bearer $GH_TOKEN" \
44            -H "Content-Type: application/json" \
45            -d "$QUERY" \
46            https://api.github.com/graphql | jq -r '.data.repository.id')
47
48          echo "repository_id=$REPO_ID" >> $GITHUB_OUTPUT
49
50      - name: Obtener fecha y hora actual
51        id: get_datetime
52        run: echo "now=$(date +%Y-%m-%dT%H:%M:%S)" >> $GITHUB_OUTPUT
53
54      - name: Crear discusión con resumen de commits
55        id: create-discussion
56        uses: abirismyname/create-discussion@main
57        with:
58          title: "Resumen de commits - ${steps.get_datetime.outputs.now}"
59          body-filepath: ${env.SUMMARY_FILE}
60          repository-id: ${steps.get_repo_id.outputs.repository_id}
61          category-id: ${secrets.DISCUSSION_CATEGORY_ID}
62          github-token: ${secrets.MY_PERSONAL_ACCESS_TOKEN}
63
64      - name: Imprimir ID y URL de Discusion
65        run: |
66          echo discussion-id: ${steps.create-discussion.outputs.discussion-id}
67          echo discussion-url: ${steps.create-discussion.outputs.discussion-url}

```

Workflow run was successfully requested.

Actions

New workflow

All workflows

Crear discusión con resumen de co...

Management

Caches

Attestations

Runners

Usage metrics

Performance metrics

Crear discusión con resumen de commits

crear-discusion-commits.yml

0 workflow runs

Event Status Branch Actor

This workflow has a workflow_dispatch event trigger.

Run workflow

Use workflow from

Branch: main

Run workflow

crear-discusion

succeeded 12 minutes ago in 5s

Search logs

- > ☒ Set up job
- > ☒ Checkout del repositorio
- > ☒ Obtener últimos 5 commits y guardar como Markdown
- > ☒ Obtener repositoryId con GraphQL
- > ☒ Obtener fecha y hora actual

☒ Crear discusión con resumen de commits

```
1 ▼ Run abirismyname/create-discussion@main
2   with:
3     title: Resumen de commits - 2025-05-20T18:23:12
4     body-filepath: RESUMEN.MD
5     repository-id: R_kgD00tXsoA
6     category-id: ***
7     github-token: ***
8     repository-name: icebeam7/Demos-Sesion-8
9     category-name: General
10  env:
11    SUMMARY_FILE: RESUMEN.MD
12  Discussion Created id: D_kwD00tXsoM4Af24w, url: https://github.com/icebeam7/discussions/7
```

- > ☒ Imprimir ID y URL de Discusion
- > ☒ Post Checkout del repositorio
- > ☒ Complete job

icebeam7 / Demos-Sesion-8

Type / to search

<> Code

Issues

Pull requests

Discussions

Actions

Projects

Wiki

Security

Insights

is:open category:"Últimos commits"

Sort by: Latest activity

Label

Filter: Open

New discussion

Categories

View all discussions

Announcements

General

Ideas

Polls

Q&A

Show and tell

Últimos commits

Últimos commits

↑ 1

Resumen de commits - 2025-05-20T18:23:12

icebeam7 started 15 minutes ago in Últimos commits

0

↑ 1

Resumen de commits - 2025-05-20T18:16:24

icebeam7 started 21 minutes ago in Últimos commits

0

↑ 1

Resumen de commits - 2025-

icebeam7 started 30 minutes ago in Últi

↑ 1

Resumen de commits - 2025-

icebeam7 started 34 minutes ago in Últi

Resumen de commits - 2025-05-20T18:23:12 #7

icebeam7 started this conversation in Últimos commits



icebeam7 15 minutes ago Maintainer

Últimos 5 commits de la rama main

- [38d1679](#) Update crear-discusion-commits.yml (Luis Beltran, 2025-05-20)
- [f14474a](#) Update crear-discusion-commits.yml (Luis Beltran, 2025-05-20)
- [66a71b3](#) Update crear-discusion-commits.yml (Luis Beltran, 2025-05-20)
- [2144015](#) Update crear-discusion-commits.yml (Luis Beltran, 2025-05-20)
- [488eb65](#) Update crear-discusion-commits.yml (Luis Beltran, 2025-05-20)

↑ 1



Q1: Who should have 2 Factor Authentication enabled on GitHub?

It's not recommended to use 2FA on GitHub for anyone

GitHub Organization owners

All GitHub users

GitHub Enterprise Server users

Autenticación a 2 pasos

Autenticación de 2 factores (2FA)

- Después de ingresar el usuario y contraseña de GitHub, con 2FA activado, se te solicitará un código adicional:
 - Generado por una aplicación de contraseña de un solo uso basada en el tiempo (TOTP)
 - Enviado mediante SMS
 - O también puedes confirmar el inicio de sesión mediante GitHub Mobile
 - Otra alternativa: llave de seguridad usando WebAuthn o Passkeys.
 - Métodos que reducen ataques basados en credenciales
 - Mejoran la UX de inicio de sesión

Two-factor authentication

Enabled

Because of your contributions on GitHub, two-factor authentication is required for your account. Thank you for helping keep the ecosystem safe! [Learn more about our two-factor authentication initiative.](#)

Two-factor authentication adds an additional layer of security to your account by requiring more than just a password to sign in. [Learn more about two-factor authentication.](#)

Preferred 2FA method

Set your preferred method to use for two-factor authentication when signing into GitHub.

Passkeys



Passkeys should be used primarily for standalone, password-less authentication.

Two-factor methods



Authenticator app Configured

Edit

Use an authentication app or browser extension to get two-factor authentication codes when prompted.



SMS/Text message

Add

Get one-time codes sent to your phone via SMS to complete authentication requests.



Security keys

Edit

Security keys are webauthn credentials that can only be used as a second factor of authentication.



GitHub Mobile Configured 1 device

Show

GitHub Mobile can be used for two-factor authentication by installing the GitHub Mobile app and signing in to your account.

Recovery options



Your two-factor authentication recovery codes have not been downloaded or printed in the last one year. Make sure your recovery codes are up-to-date by viewing and downloading or printing them again.



Recovery codes Viewed

View

Recovery codes can be used to access your account in the event you lose access to your device and cannot receive two-factor authentication codes.

Q1: What are the different forms of two-factor or multi-factor authentication supported by GitHub? (Choose five.)

Security key

Time-based one-time password (TOTP)

Text message

Phone call

GitHub mobile

Passkey

Email

🔔 Notifications

Access

📁 Billing and licensing

✉ Emails

🛡️ Password and authentication

👤 Sessions

🔑 SSH and GPG keys

🏢 Organizations

🌐 Enterprises

⚠ Moderation

Code, planning, and automation

💻 Repositories

💻 Codespaces

📦 Packages

🤖 Copilot

📄 Pages

← Saved replies

Security

Passkeys

Passkeys are webauthn credentials that validate your identity using touch, facial recognition, a device password, or a PIN. They can be used as a password replacement or as a 2FA method. [Learn more about passkeys.](#)

Your passkeys

Add a passkey



GitHubLuis

Seen from this browser

Synced

Added on Jul 1, 2024 | Last used 2 weeks ago



Two-factor authentication

Enabled



Because of your contributions on GitHub, two-factor authentication is required for your account. Thank you for helping keep the ecosystem safe! [Learn more about our two-factor authentication initiative.](#)

Two-factor authentication adds an additional layer of security to your account by requiring more than just a password to sign in. [Learn more about two-factor authentication.](#)

Preferred 2FA method

Set your preferred method to use for two-factor authentication when signing into GitHub.

Passkeys



Passkeys should be used primarily for standalone, password-less authentication.

Passkey (clave de acceso)

- Permite habilitar un inicio de sesión seguro y sin código adicional.
- Un passkey satisface los requisitos de contraseña y 2FA, por lo que puede completar su inicio de sesión con un solo paso.
- Se enlaza a un dispositivo físico, es resistente al phishing.



Configure passwordless authentication



Add a passkey

Your device supports passkeys, a password replacement that validates your identity using touch, facial recognition, a device password, or a PIN.

Passkeys can be used for sign-in as a simple and secure alternative to your password and two-factor credentials.

[Add passkey](#)

[Ask me later](#)

Not interested?

[Don't ask again for this browser](#)

- Passkeys: pares de claves criptográficas (una clave pública y una clave privada) almacenadas por un autenticador controlado por el usuario.
- El autenticador puede demostrar que un usuario está presente y está autorizado a usar el passkey.
- Los autenticadores prueban la autorización con un PIN, un código de acceso, una biométrica o una contraseña de dispositivo, dependiendo de las funcionalidades y la configuración del autenticador.
- Los autenticadores vienen en muchas formas, como un dispositivo iPhone o Android, Windows Hello, una clave de seguridad de hardware FIDO2 o un administrador de contraseñas.



Two-factor authentication



Passkey

When you are ready, authenticate using the button below.

[Use passkey](#)

Unable to verify with your passkey?

- [Use GitHub Mobile](#)
- [Use your authenticator app](#)
- [Use a recovery code or begin 2FA account recovery](#)



Use a passkey from Redmi Note 9 Pro to sign in to github.com

[Use a different passkey](#)

[Cancel](#)

[Continue](#)



Check your device

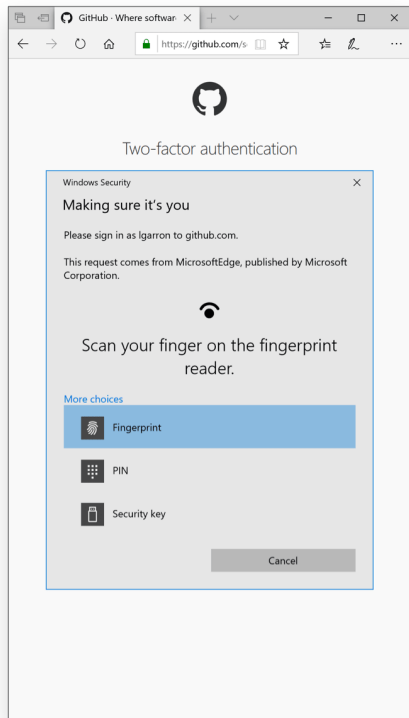
A notification was sent to Redmi Note 9 Pro

[Cancel](#)

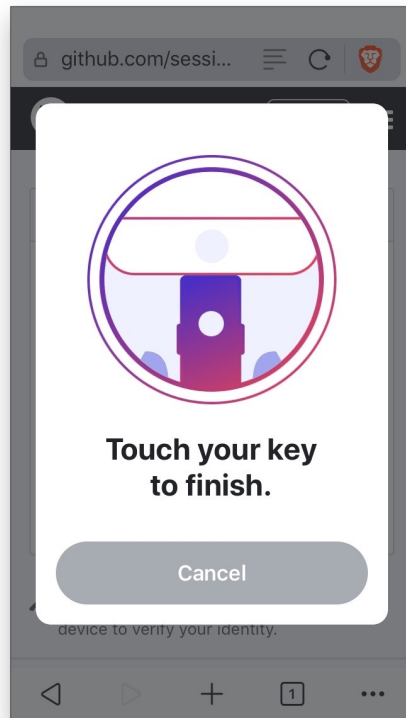
WebAuthn (security keys)

- Soporta elementos biométricos y hardware tokens como YubiKey

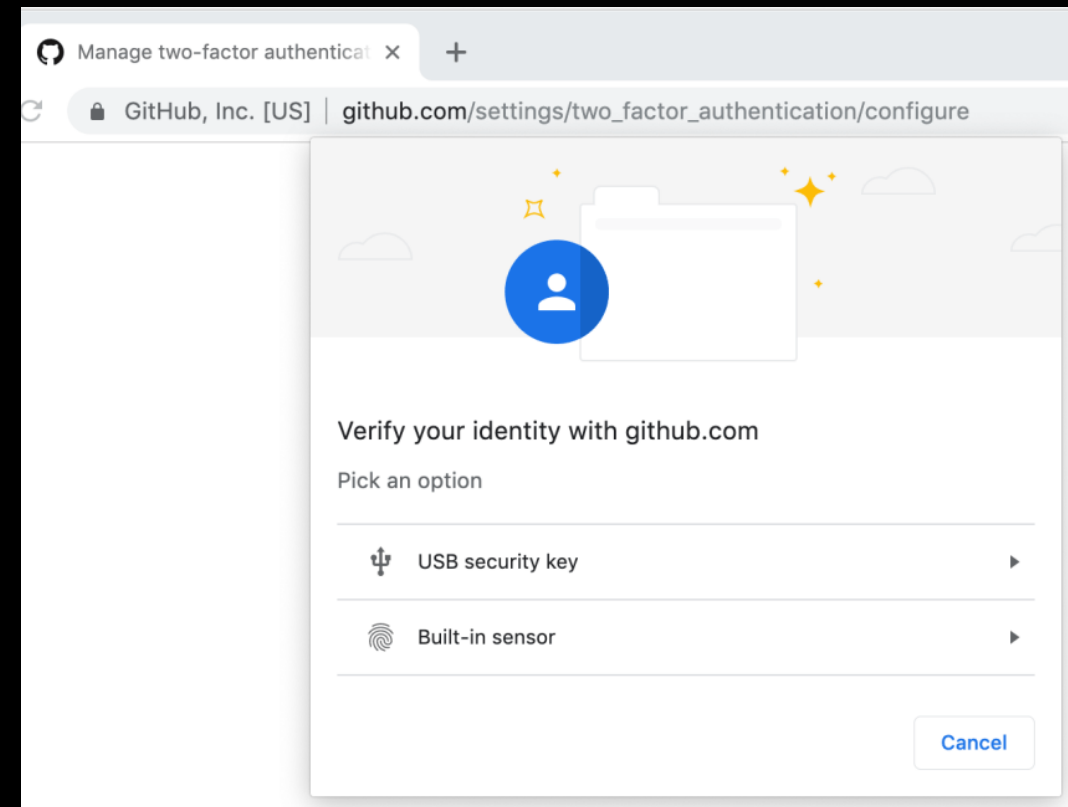
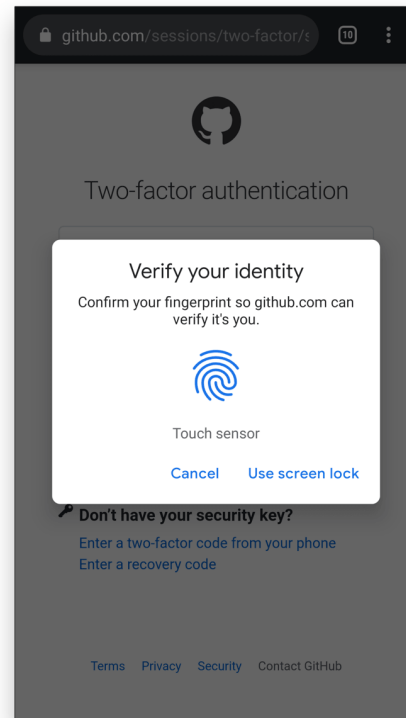
Windows MS Edge



iOS Brave

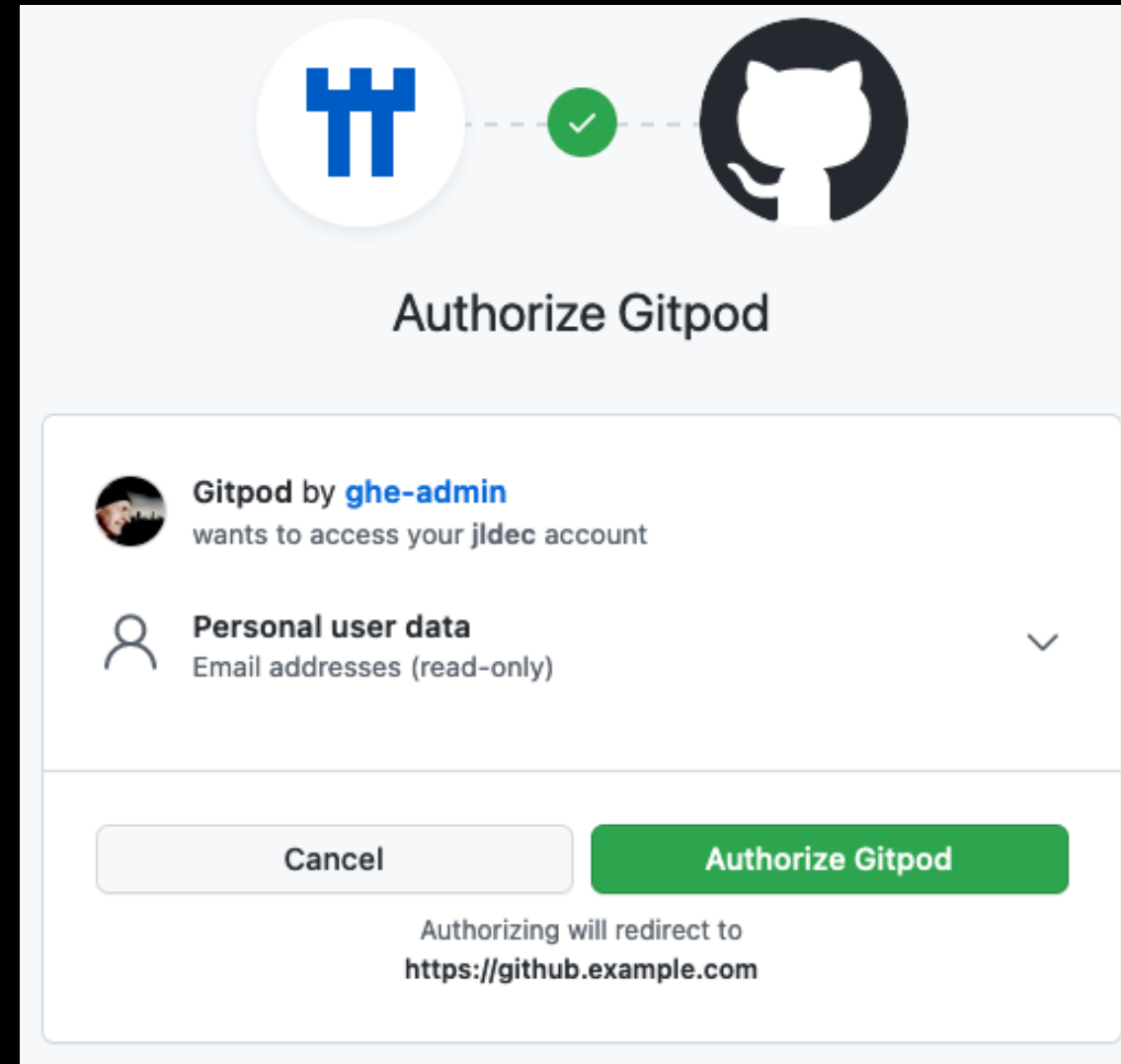


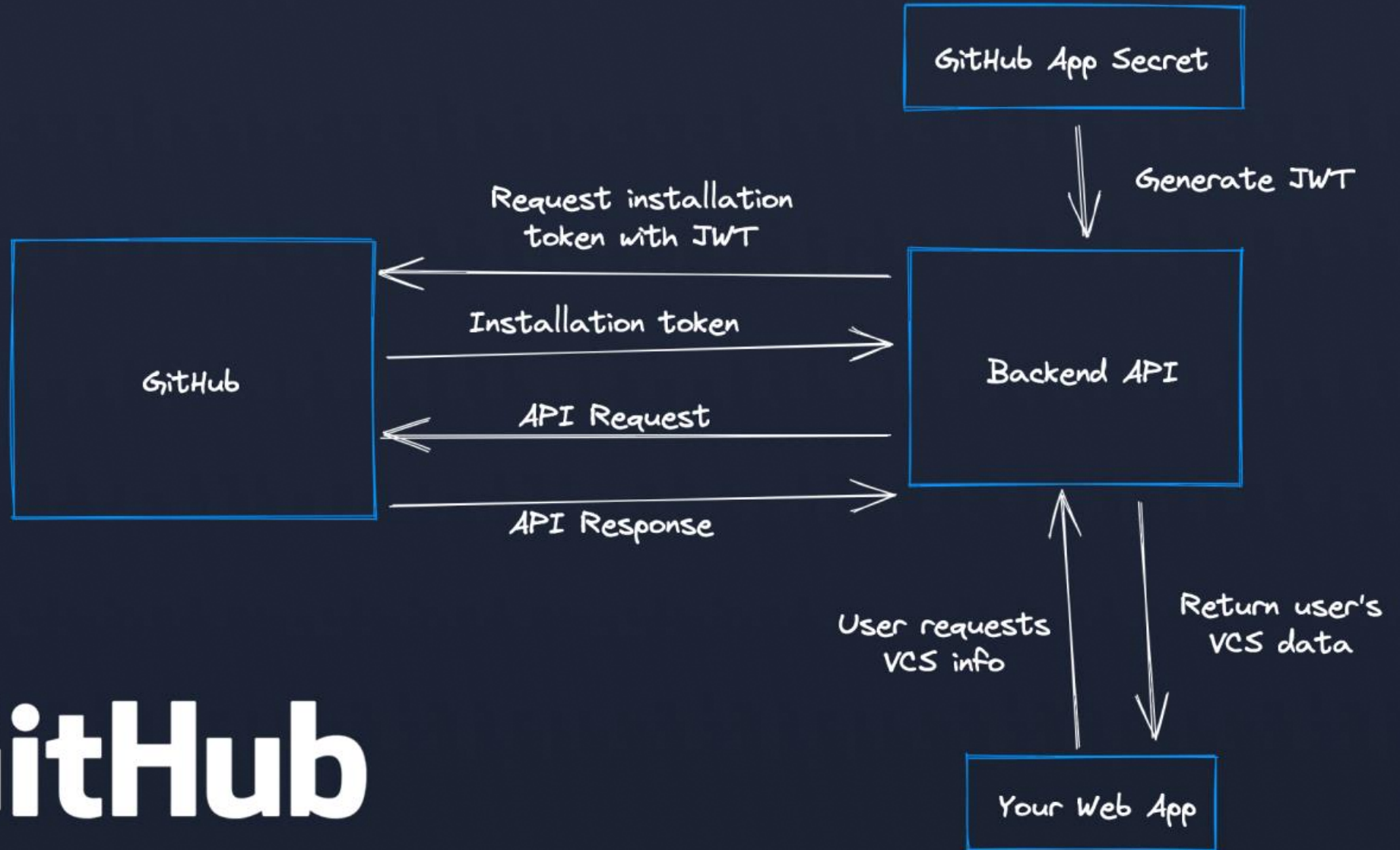
Android Chrome



Otros mecanismos

- Las aplicaciones de OAuth usan el flujo de OAuth 2.0 de GitHub para autenticar a los usuarios y conceder acceso con ámbito a aplicaciones externas.
- Las GitHub Apps se autentican como instalaciones individuales con permisos detallados y son ideales para CI/CD y pipelines de automatización.





GitHub App and OAuth Integration

Claves SSH

- Eliminan la necesidad de username + PAT.
- Al configurar SSH:
 - Se genera una clave SSH
 - Se agrega al agente de SSH (nivel de seguridad adicional con un passkey)
 - Se agrega a su cuenta de GitHub. La adición de la clave SSH al agente de SSH garantiza que la clave SSH tenga un nivel de seguridad adicional mediante el uso de un passphrase.
 - Se puede configurar la copia local de Git para interactuar con GitHub.

Demo 3 – Commit via SSH

```
[luisbeltran@Luiss-MacBook-Pro-2 ~ % cd ~/.ssh
[luisbeltran@Luiss-MacBook-Pro-2 .ssh % ssh-keygen -t ed25519 -C "luis@luisbeltran.mx"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/Users/luisbeltran/.ssh/id_ed25519):
/Users/luisbeltran/.ssh/id_ed25519 already exists.
Overwrite (y/n)? y
[Enter passphrase (empty for no passphrase):
[Enter same passphrase again:
Your identification has been saved in /Users/luisbeltran/.ssh/id_ed25519
Your public key has been saved in /Users/luisbeltran/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:ziMYDc73s9vRgU9dqBFFx3gqFrQBvf+iiAHncrVgk40 luis@luisbeltran.mx
The key's randomart image is:
+--[ED25519 256]--+
|                .+0  +. |
|                ooo =|
|      .          .+ =. |
|    o o    +  .+.*. |
```

<https://docs.github.com/en/authentication/managing-commit-signature-verification/about-commit-signature-verification#ssh-commit-signature-verification>

```
[luisbeltran@Luiss-MacBook-Pro-2 ~ % eval "$(ssh-agent -s)"
Agent pid 77727
[luisbeltran@Luiss-MacBook-Pro-2 ~ % open ~/.ssh/config
The file /Users/luisbeltran/.ssh/config does not exist.
[luisbeltran@Luiss-MacBook-Pro-2 ~ % touch ~/.ssh/config
[luisbeltran@Luiss-MacBook-Pro-2 ~ % code ~/.ssh/config
[luisbeltran@Luiss-MacBook-Pro-2 ~ % ssh-add --apple-use-keychain ~/.ssh/id_ed25519
Enter passphrase for /Users/luisbeltran/.ssh/id_ed25519:
Identity added: /Users/luisbeltran/.ssh/id_ed25519 (luis@luisbeltran.mx)
[luisbeltran@Luiss-MacBook-Pro-2 ~ % pbcopy < ~/.ssh/id_ed25519.pub
```

⬇ config

×

Users > luisbeltran > .ssh > ⬇ config

1	Host github.com
2	AddKeysToAgent yes
3	UseKeychain yes
4	IdentityFile ~/.ssh/id_ed25519

```

luisbeltran@Luiss-MacBook-Pro-2 .ssh % ssh -vT git@github.com
OpenSSH_9.0p1, LibreSSL 3.3.6
debug1: Reading configuration data /Users/luisbeltran/.ssh/config
debug1: /Users/luisbeltran/.ssh/config line 1: Applying options for github.com
debug1: Reading configuration data /etc/ssh/ssh_config
debug1: /etc/ssh/ssh_config line 21: include /etc/ssh/ssh_config.d/* matched no files
debug1: /etc/ssh/ssh_config line 54: Applying options for *
debug1: Authenticator provider $SSH_SK_PROVIDER did not resolve; disabling
debug1: Connecting to github.com port 22.
debug1: Connection established.
debug1: identity file /Users/luisbeltran/.ssh/id_ed25519 type 3
debug1: identity file /Users/luisbeltran/.ssh/id_ed25519-cert type -1
debug1: Local version string SSH-2.0-OpenSSH_9.0
debug1: Remote protocol version 2.0, remote software version 4c545346
debug1: compat_banner: no match: 4c545346 debug1: Offering public key: /Users/luisbeltran/.ssh/id_ed25519 ED25519 SHA256:ziMYDc73s9vRgU9dqBFFx
debug1: Authenticating to github.com:22 as 'git' debug1: Server accepts key: /Users/luisbeltran/.ssh/id_ed25519 ED25519 SHA256:ziMYDc73s9vRgU9dqBFFx3
debug1: load_hostkeys: fopen /Users/luisbeltran/.ssh/known_hosts: No such file or directory
debug1: load_hostkeys: fopen /etc/ssh/ssh_known_hosts: No such file or directory
debug1: load_hostkeys: fopen /etc/ssh/ssh_known_hosts2: No such file or directory
debug1: channel 0: new [client-session]
debug1: Entering interactive session.
debug1: SSH2_MSG_KEXINIT sent debug1: pldge: filesystem
debug1: SSH2_MSG_KEXINIT received debug1: client_input_global_request: rtype hostkeys-00@openssh.com want_reply 0
debug1: kex: algorithm: curve25519-sha256 debug1: client_input_hostkeys: searching /Users/luisbeltran/.ssh/known_hosts for github.com / (none)
debug1: kex: host key algorithm: ssh-ed25519 debug1: client_input_hostkeys: searching /Users/luisbeltran/.ssh/known_hosts2 for github.com / (none)
debug1: kex: server->client cipher: chacha20-poly1305 MAC: hmac-sha2-256 debug1: client_input_hostkeys: hostkeys file /Users/luisbeltran/.ssh/known_hosts2 does not exist
debug1: Sending environment.
debug1: channel 0: setting env LC_CTYPE = "UTF-8"
debug1: channel 0: setting env LANG = "en_US.UTF-8"
debug1: channel 0: setting env GIT_SSH_COMMAND = "ssh -vT"
debug1: channel 0: setting env GIT_SSH = "ssh -vT"
debug1: channel 0: setting env GIT_SSH_OPTS = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS2 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS3 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS4 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS5 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS6 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS7 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS8 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS9 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS10 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS11 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS12 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS13 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS14 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS15 = "-vT"
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debug1: channel 0: setting env GIT_SSH_ARGS24 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS25 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS26 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS27 = "-vT"
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debug1: channel 0: setting env GIT_SSH_ARGS30 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS31 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS32 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS33 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS34 = "-vT"
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debug1: channel 0: setting env GIT_SSH_ARGS37 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS38 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS39 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS40 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS41 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS42 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS43 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS44 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS45 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS46 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS47 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS48 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS49 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS50 = "-vT"
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debug1: channel 0: setting env GIT_SSH_ARGS61 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS62 = "-vT"
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debug1: channel 0: setting env GIT_SSH_ARGS153 = "-vT"
debug1: channel 0: setting env GIT_SSH_ARGS154 = "-vT"

```

 **Demos-Sesion-8** Public

 Pin  Unwatch 1

 main

 1 Branch

 0 Tags

 t

Add file

 Code

 **icebeam7** Update crear-discusion-commits.yml

 .github/workflows Update crear-discu

 LICENSE Initial commit

 README.md Initial commit

 README

 MIT license

Demos-Sesion-8

Local

Codespaces

Copilot

 Clone 

HTTPS

SSH

GitHub CLI

git@github.com:icebeam7/Demos-Sesion-8.git



Use a password-protected SSH key.

 Open with GitHub Desktop

Open with Visual Studio

 Download ZIP

```
[luisbeltran@Luiss-MacBook-Pro-2 .ssh % git config --global gpg.format ssh
[luisbeltran@Luiss-MacBook-Pro-2 .ssh % git config --global user.signingkey ~/.ssh/id_ed25519.pub
[luisbeltran@Luiss-MacBook-Pro-2 .ssh % cd ..
```



```

[luibeltran@Luiss-MacBook-Pro-2 ~ % git clone git@github.com:icebeam7/Demos-Sesion-8.git
Cloning into 'Demos-Sesion-8'...
remote: Enumerating objects: 94, done.
remote: Counting objects: 100% (94/94), done.
remote: Compressing objects: 100% (76/76), done.
remote: Total 94 (delta 17), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (94/94), 22.75 KiB | 253.00 KiB/s, done.
Resolving deltas: 100% (17/17), done.
[luibeltran@Luiss-MacBook-Pro-2 ~ % cd Demos-Sesion-8
[luibeltran@Luiss-MacBook-Pro-2 Demos-Sesion-8 % touch newfile.txt
[luibeltran@Luiss-MacBook-Pro-2 Demos-Sesion-8 % git add -A
[luibeltran@Luiss-MacBook-Pro-2 Demos-Sesion-8 % git commit -am "Add new text file"
[main b0fe461] Add new text file
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 newfile.txt
[luibeltran@Luiss-MacBook-Pro-2 Demos-Sesion-8 % git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 281 bytes | 281.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 commit.
To github.com:icebeam7/Demos-Sesion-8.git
38d1679..b0fe461 main -> main
[luibeltran@Luiss-MacBook-Pro-2 Demos-Sesion-8 % █

```


Demos-Sesion-8
Public
Pin
Unwatch 1

main
1 Branch
0 Tags

Add file
Code

 icebeam7	Add new text file	b0fe461 · 1 minute ago	20 Commits
 .github/workflows	Update crear-discusion-commits.yml	3 hours ago	
 LICENSE	Initial commit	7 hours ago	
 README.md	Initial commit	7 hours ago	
 newfile.txt	Add new text file	1 minute ago	

README
MIT license

 **Luis Beltran (icebeam7)**
Your personal account [Switch settings context](#) ▼

[Go to your personal profile](#)

- Public profile
- Account
- Appearance
- Accessibility
- Notifications
- Access
 - Billing and licensing
 - Emails
 - Password and authentication
 - Sessions
 - SSH and GPG keys**
 - Organizations
 - Enterprises
 - Moderation

SSH keys

New SSH key

There are no SSH keys associated with your account.
Check out our guide to [connecting to GitHub using SSH keys](#) or trou

GPG keys

There are no GPG keys associated with your account.
Learn how to [generate a GPG key and add it to your account](#).

Vigilant mode

- ☐ **Flag unsigned commits as unverified**
This will include any commit attributed to your account but not s
Note that this will include your existing unsigned commits.
[Learn about vigilant mode.](#)

Add new SSH Key

Title

My Mac Key

Key type

Authentication Key

Key

ssh-ed25519 AAAAC3NzaC1lZDI1N' [REDACTED] luis@luisbeltran.mx

Add SSH key

SSH keys

This is a list of SSH keys associated with your account. Remove any keys that you do not r

Authentication keys



SSH

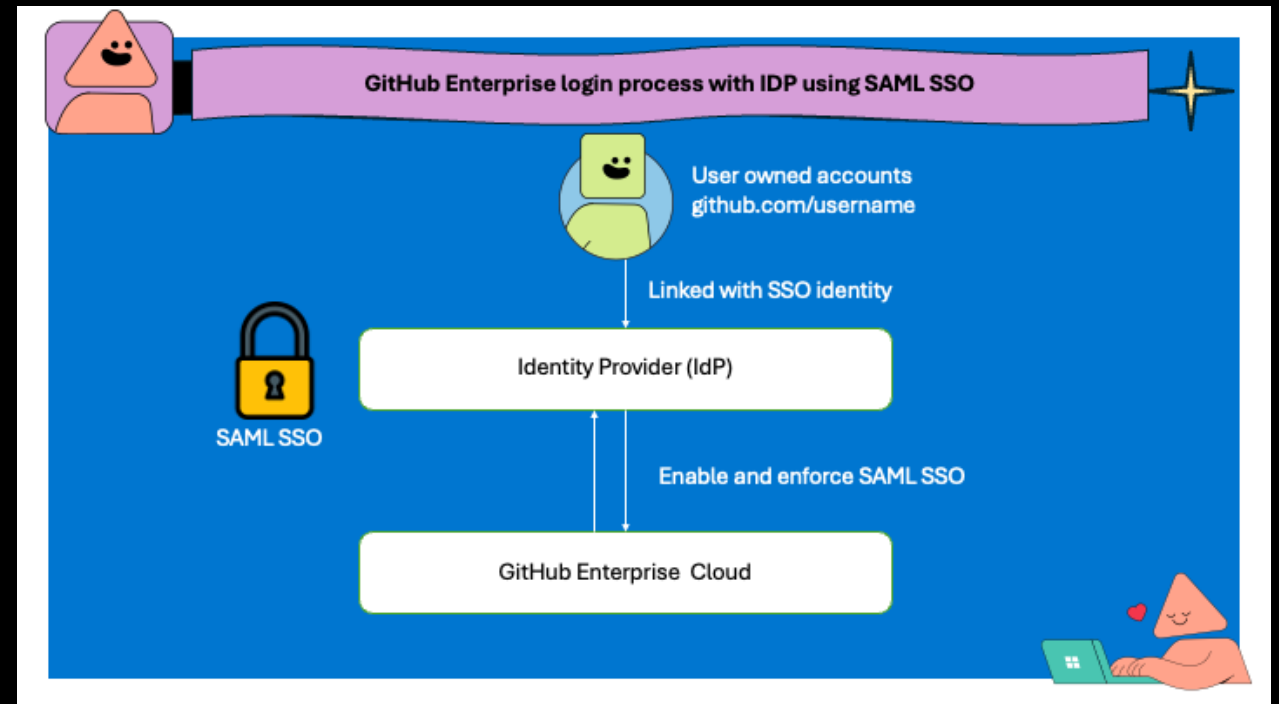
My Mac Key
SHA256 : ziMYDc73s!
Added on May 20, 2025
Never used — Read/write

SAML SSO (single sign-on)

- Permite el acceso a recursos que pertenecen a una organización o cuenta empresarial que utiliza el inicio de sesión único de SAML
- Es posible que también debas autenticarte a través de un Proveedor de Identidad (IdP)

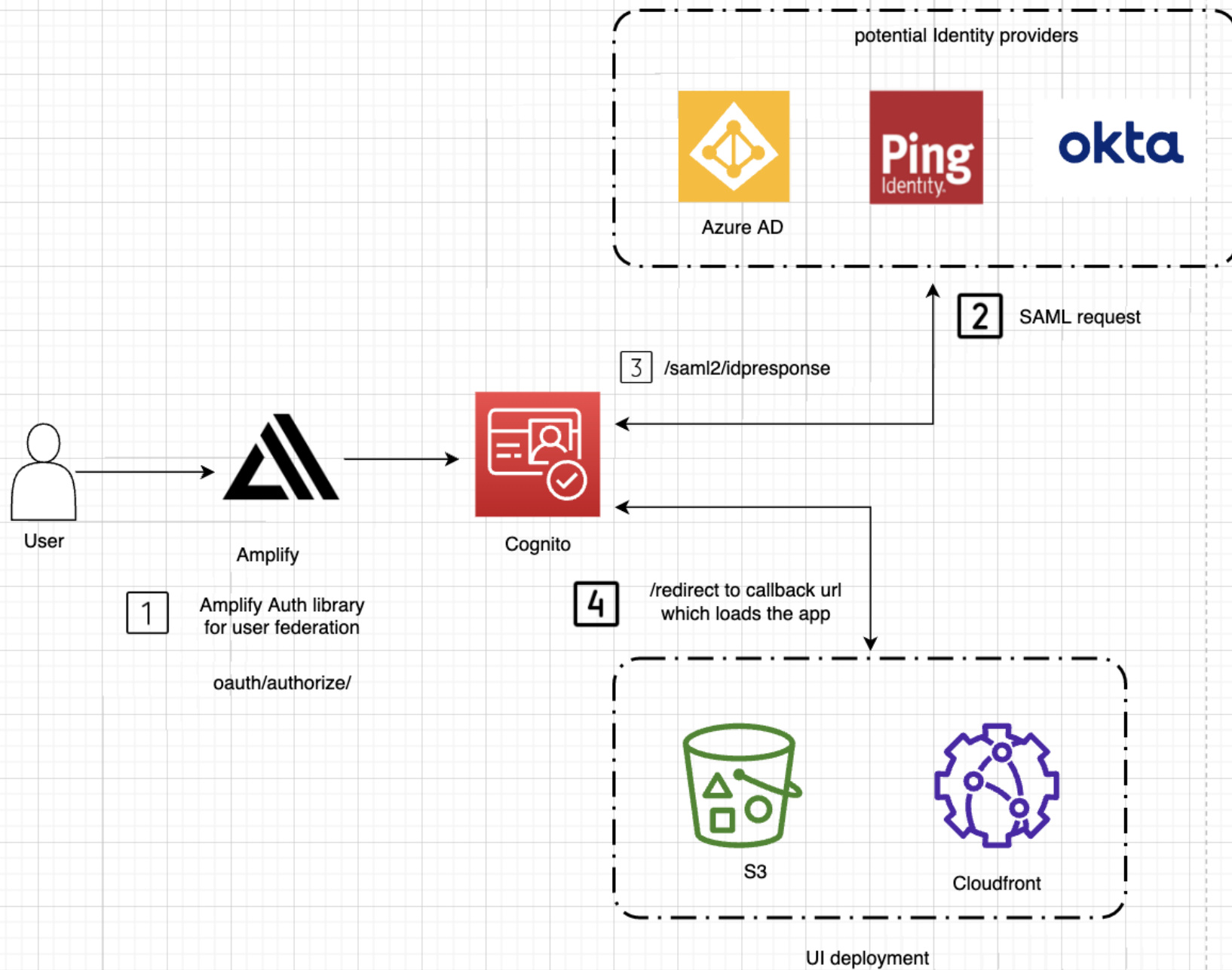
Se puede usar claves SSH con un repositorio que sea propiedad de una organización que use el inicio de sesión único (SSO) de SAML.

Si la organización proporciona certificados SSH, los usuarios también pueden usarlos para acceder a los repositorios de la organización sin agregar el certificado a su cuenta de GitHub.



GitHub ofrece soporte limitado para todos los proveedores de identidad que implementan el estándar SAML 2.0 con soporte oficial para varios proveedores de identidad populares, entre ellos:

- Servicios de federación de Active Directory (AD FS).
- Microsoft Entra ID
- Okta
- OneLogin
- PingOne



Security

Audit log

Blocked users

Webhooks

Third-party access

Installed integrations

Repository topics

Developer settings

OAuth applications

Integrations

SAML single sign-on

Manage your organization's membership while adding another level of security with SAML. [Learn more](#)

☒ Enable SAML authentication

Enable SAML authentication for your organization through an identity provider like Azure, Okta, Onelogin, Ping Identity or your custom SAML 2.0 provider.

Sign on URL

Members will be forwarded here when signing in to your organization

Issuer

Typically a unique URL generated by your SAML Identity Provider

Public certificate

Paste your x509 certificate here

Organization

Switch settings context

Go to your organization profile

General

Policies

Access

Billing and plans

Organization roles

Repository roles

Member privileges

Import/Export

Moderation

Code, planning, and automation

Repository

Codespaces

Planning

Copilot

Actions

Models

Webhooks

Discussions

Packages

Pages

Security

Authentication security

Advanced Security

Deploy keys

Compliance

Verified and approved domains

Secrets and variables

Two-factor authentication

Two-factor authentication adds another level of security for your organization. [Learn more about requiring two-factor authentication in your organization.](#)

☐

Require two-factor authentication for everyone in the WhatTheHack-Código Facilito organization.

Organization members who do not have two-factor authentication enabled will be unable to access resources owned by the WhatTheHack-Código Facilito organization, but will remain a member of WhatTheHack-Código Facilito until they update their settings. Outside collaborators who do not have two-factor authentication enabled will be removed from the organization and notified. [View organization membership](#) to see which users will be impacted.

Save

Secure your data and people with GitHub Enterprise

Start free for 30 days

SAML single sign-on

SAML single sign-on (SSO) gives GitHub Enterprise accounts a way to control and secure access to organization resources like repositories, issues, and pull requests. [Learn more](#)

SSH certificate authorities

With an SSH certificate authority, enterprise accounts can provide SSH certificates that members can use to access your resources with Git. [Learn more](#)

Certified for compliance

GitHub Enterprise Cloud is certified for SOC 1 Type 2, SOC 2 Type 2, FedRAMP LiSaaS. Enterprise customers can download these reports, and more, at any time. [Learn more](#)

Team Synchronization

Enable and disable team synchronization between your identity provider (IdP) and your organizations on GitHub Enterprise accounts. [Learn more](#)

IP allow list

GitHub Enterprise accounts can restrict access to private organization assets by configuring an allow list for specific IP addresses. [Learn more](#)

Enterprise Managed Users

Q1: What are Enterprise Managed Users?

It's an user account without access to GitHub web UI but is rather only used for automation tasks in your GitHub Enterprise

It's a GitHub Enterprise support user that can be used to manage your Enterprise by GitHub Support once requested

It's a GitHub Enterprise feature that allows you to manage user access from an external identity provider such as Azure AD or Okta

It's a shared user account that can be used by multiple users in your Enterprise

Gestión de permisos

Gestión de permisos

- Controla **quién puede ver o modificar** los repositorios y recursos dentro de una cuenta o una organización.
- Niveles de permisos en un repositorio:
 - Read: Ver código, issues y pull requests.
 - Triage: Clasificar issues y PRs sin modificar el código.
 - Write: Subir commits, manejar ramas, cerrar issues.
 - Maintain: Gestionar configuración del repo sin control total.
 - Admin: Control completo (incluye eliminar el repo o cambiar la visibilidad).
- Las organizaciones pueden usar **Teams** y **Roles personalizados** para gestionar permisos a escala.



General



Policies

Access

Billing and plans

Organization roles

Role management

Role assignments

Repository roles

Member privileges

Import/Export

Moderation

Code, planning, and automation

Repository

Codespaces

Planning

Copilot

Actions

Models

Webhooks

Preview

Role management

Organization roles are used to grant access to subsets of organization settings to teams and members.



All-repository read

Grants read access to all repositories in the organization.



All-repository write

Grants write access to all repositories in the organization.



All-repository triage

Grants triage access to all repositories in the organization.



All-repository maintain

Grants maintenance access to all repositories in the organization.



All-repository admin

Grants admin access to all repositories in the organization.



CI/CD Admin

Grants admin access to manage Actions policies, runners, runner groups, network configurations, secrets, variables, and usage metrics for an organization.



Security manager

Grants the ability to manage security policies, security alerts, and security configurations for an organization and all its repositories.





All-repository write

Grants write access to all repositories in the organization.



Repository

Base repository role: **Write**

Issue and Pull Request

- Assign or remove a user
- Add or remove a label
- Remove an assigned user
- Remove a label

Issue

- Close an issue
- Mark an issue as a duplicate
- Reopen a closed issue
- Set an issue type

Pull Request

- Close a pull request
- Reopen a closed pull request
- Request a pull request review

Repository

- Set milestones

Security

- View code scanning alerts
- Dismiss or reopen Dependabot alerts
- View Dependabot alerts
- Dismiss or reopen code scanning alerts

Overview Repositories Projects Packages Teams People Insights Settings



WhatTheHack-Código Facilito

Organization [Switch settings context](#)

General

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Actions

Assign

Search

Users



Assign teams or users for an organization role in WhatTheHack-CF

Search for users or teams

Choose a Role

[View role details](#)

- ☐ **All-repository read**
Grants read access to all repositories in the organization.
- ☐ **All-repository write**
Grants write access to all repositories in the organization.
- ☐ **All-repository triage**
Grants triage access to all repositories in the organization.
- ☐ **All-repository maintain**
Grants maintenance access to all repositories in the organization.
- ☐ **All-repository admin**
Grants admin access to all repositories in the organization.
- ☐ **CI/CD Admin**
Grants admin access to manage Actions policies, runners, runner groups, network configurations, secrets, variables, and usage metrics for an

Add new assignment

[Go to your organization profile](#)

Role: All

New role assignment

1 role

Visibilidad de repositorios

Visibilidad de repositorios

Determina **quién puede acceder** a un repositorio.

- Tipos:
 - Público: Visible para todos en Internet. Ideal para proyectos open source.
 - Privado: Solo accesible para personas con permiso. Recomendado para proyectos internos o confidenciales.
 - Interno (solo en GitHub Enterprise): Visible para cualquier miembro de la organización.
- Puedes cambiar la visibilidad en cualquier momento desde la configuración del repositorio.

Q1: Which of these can make a repository more discoverable? (Choose three.)

Setting up topics for the repository

A well-crafted collection of markdown documents like `README.md` , `CONTRIBUTING.md` and others.

Giving the repository a descriptive name

Using a wide range of GitHub Apps and GitHub Actions

Defining proper CI/CD pipelines

Covering the repository software with tests

Opciones de la pestaña de seguridad

Mantener un repositorio seguro

 icebeam7 / Demos-Sesion-8

🔍 Type  to search

 Code  Issues  Pull requests  Discussions  Actions  Projects  Wiki  **Security**  Insights  Settings

 Overview

Reporting

 Policy

 Advisories

Vulnerability alerts

 Dependabot

 Code scanning

 Secret scanning

Security overview

Security policy • Disabled
Define how users should report security vulnerabilities for this repository

Set up a security policy

Security advisories • Enabled
View or disclose security advisories for this repository

[View security advisories](#)

Private vulnerability reporting • Disabled
Allow users to privately report potential security vulnerabilities

Enable vulnerability reporting

Dependabot alerts • Disabled
Get notified when one of your dependencies has a vulnerability

Enable Dependabot alerts

Code scanning alerts • Needs setup
Automatically detect common vulnerability and coding errors

Set up code scanning

Secret scanning alerts • Enabled
Get notified when a secret is pushed to this repository

[View detected secrets](#)

Características de la pestaña Seguridad

- **Directivas de seguridad:** permiten especificar cómo notificar una vulnerabilidad de seguridad en el proyecto agregando un archivo SECURITY.md al repositorio.
- **Alertas de Dependabot:** le notifican cuando GitHub detecta que el repositorio usa una dependencia vulnerable o malware.
- **Avisos de seguridad:** para analizar, corregir y publicar información sobre vulnerabilidades de seguridad en el repositorio de forma privada.
- **Code Scanning:** le ayuda a encontrar, evaluar y corregir vulnerabilidades y errores en el código.

Directivas de Seguridad

- Las ventajas de la comunidad de GitHub son considerables, pero también conllevan posibles riesgos.
- El hecho de que cualquiera pueda proponer correcciones de errores públicamente comporta ciertas responsabilidades.
- La más importante es la revelación responsable de la información que podría dar lugar a vulnerabilidades de seguridad antes de que se pudieran solucionar los errores subyacentes.
- Los desarrolladores que intentan comunicar o abordar problemas de seguridad buscan un archivo SECURITY.md en la raíz de un repositorio para revelar de forma responsable sus inquietudes.
- Al proporcionar instrucciones en este archivo, se acelera la resolución de estos problemas críticos.



📄 Overview

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📦 Dependabot

🔍 Code scanning

🔑 Secret scanning

SECURITY.md



Política de Seguridad

Gracias por tu interés en mejorar la seguridad de este proyecto .NET.

Este documento describe cómo reportar vulnerabilidades de manera responsable.

Cómo Reportar Vulnerabilidades

Si descubres una vulnerabilidad en este proyecto, **no abras un issue público**. En su lugar, sigue este proceso:

1. Envía un correo a [\[security@example.com\]](mailto:security@example.com) con:
 - Descripción detallada del problema
 - Pasos para reproducirlo (incluyendo código de ejemplo, si es posible)
 - Posible impacto
 - Archivos afectados o paquetes NuGet relacionados
 - Tu información de contacto (opcional)

Nos comprometemos a responder lo antes posible y trabajar contigo de forma confidencial.

Alcance

Esta política cubre vulnerabilidades en el código fuente de este repositorio, incluidas:

- Aplicaciones ASP.NET Core o Blazor
- Librerías en C# (.NET Standard/.NET Core/.NET 6+)

Security Advisories

- Los avisos de seguridad de GitHub permiten a los mantenedores de repositorios analizar y corregir de forma privada una vulnerabilidad de seguridad en un proyecto.
- Después de que los mantenedores de repositorios colaboren en una corrección, pueden publicar el aviso de seguridad para revelar públicamente la vulnerabilidad de seguridad a la comunidad del proyecto.
- Al publicar avisos de seguridad, los mantenedores de repositorios facilitan a su comunidad la actualización de dependencias de paquetes y la investigación de las consecuencias de las vulnerabilidades de seguridad.
- GitHub almacena los avisos publicados en la lista Vulnerabilidades y exposiciones comunes (CVE). Esta lista se usa para notificar automáticamente a los repositorios afectados que usen software que tenga una vulnerabilidad listada.

Open a draft security advisory

After the draft security advisory is open, you can privately discuss it with collaborators and create a temporary private fork where you can collaborate on the vulnerability, just fill out the draft security advisory and then publish it.



Title

Affected version(s)

Patched version(s)

Package name(s)

Package ecosystem(s)

Severity

CVE identifier

Write

Preview

H B I

≡ <> 🔗

≡ ≡ ≡

☑

@

🔗

↩

Impact

What kind of vulnerability is it? Who is impacted?

Patches

Has the problem been patched? What versions should users upgrade to?

Workarounds

Is there a way for users to fix or remediate the vulnerability without upgrading?

References

Are there any links users can visit to find out more?

For more information

If you have any questions or comments about this advisory:

* Open an issue in [example link to repo](http://example.com)

* Email us at [example email address](mailto:example@example.com)

Attach files by dragging & dropping, selecting or pasting them.

Styling with Markdown is supported

Create security advisory

Access and vis

Until it is published, the advisory will only be shared with collaborators within the department-of-affairs/supreme-court teams within the state. It will be added once the

Once published in a
private repository
everyone who

They are not b
Advisory Datab
and receive a C

- Prácticamente todos los proyectos hoy en día tienen dependencias en paquetes externos.
- Aunque estos componentes pueden ofrecer ventajas considerables en cuanto a productividad, pueden introducir otros riesgos de seguridad.
- Estar atento a estos paquetes y su estado de vulnerabilidad puede llevar mucho tiempo, especialmente porque cada dependencia podría tener sus propias dependencias, lo que dificulta su seguimiento y mantenimiento.
- GitHub proporciona características que reducen esta carga de trabajo.

Gráficos de dependencias de repositorios

- Una característica predeterminada que incluyen todos los repositorios son los gráficos de dependencias.
- GitHub examina los manifiestos de paquete comunes, como `package.json`, `requirements.txt` y otros.
- Estos gráficos permiten a los propietarios de proyectos realizar un seguimiento recurrente de todas las dependencias en las que se basa su proyecto.

Dependency graph

Dependencies

Dependents

Dependabot

 Export SBOM

 Search all dependencies

 3 Total	
actions/checkout 3	Detected automatically on Feb 05, 2024 (GitHub Actions) · .github/workflows/workflow.yml
azure/arm-deploy 1	Detected automatically on Feb 05, 2024 (GitHub Actions) · .github/workflows/workflow.yml
azure/login 1	Detected automatically on Feb 05, 2024 (GitHub Actions) · .github/workflows/workflow.yml

Alertas de Dependabot

- Incluso con un gráfico de dependencias visual, puede ser abrumador estar atento a las consideraciones de seguridad más recientes de cada una de las dependencias de un proyecto.
- Para reducir esta sobrecarga, GitHub proporciona alertas de Dependabot que vigilan los gráficos de dependencias automáticamente.
- Luego, comparan las versiones de destino con las versiones de listas de vulnerabilidades conocidas.
- Cuando se detecta un riesgo, se alerta al proyecto.
- La entrada para el análisis procede de avisos de seguridad de GitHub.

Dependabot alerts

Configure ▾

🔍 is:open

☐  5 Open ☒ 0 Closed

Package ▾ Ecosystem ▾ Manifest ▾ Severity ▾ Sort ▾

☐  **Command Injection in hot-formula-parser** Critical
#1 opened now • Detected in hot-formula-parser (npm) • javascript/package-lock.json

☐  **Command Injection in lodash** High
#5 opened now • Detected in lodash (npm) • javascript/yarn.lock

☐  **Command Injection in lodash** High
#3 opened now • Detected in lodash (npm) • javascript/package-lock.json

☐  **Regular Expression Denial of Service (ReDoS) in lodash** Moderate
#4 opened now • Detected in lodash (npm) • javascript/yarn.lock

☐  **Regular Expression Denial of Service (ReDoS) in lodash** Moderate
#2 opened now • Detected in lodash (npm) • javascript/package-lock.json

Secret Scanning

- Detecta **credenciales expuestas accidentalmente** en el código fuente, como claves API, tokens de acceso, contraseñas, etc.
- Escanea los commits, ramas y pull requests en busca de patrones conocidos de **secretos** (por ejemplo, tokens de AWS, claves de Google Cloud, etc.).
- Puede escanear tanto **repos públicos como privados**.
- Cuando se encuentra un secreto, GitHub puede:
 - Notificar al autor del commit y/o al equipo de seguridad.
 - En algunos casos, revocar automáticamente el secreto (según el proveedor).
 - Integrarse con herramientas personalizadas para manejar alertas



Demos-Sesion-8 /

newfile.txt

in main

Edit

Preview

1

ghp_VpYeEj [REDACTED]

[Secret scanning](#) found a **GitHub Personal Access Token** secret on [line 1](#).

Allowing this secret risks exposure. Instead, consider [removing the secret from your commit and commit history](#).

Exposing this secret can allow someone to:

- Verify the identity of this **GitHub Personal Access Token** secret
- Know which resources this secret can access
- Act on behalf of the secret's owner
- Push this secret to this repository without being blocked

☐ It's used in tests

The secret poses no risk. If anyone finds it, they cannot do any damage or gain access to sensitive information.

☐ It's a false positive

The detected string is not a secret.

☐ I'll fix it later

The secret is real, I **understand the risk**, and I will need to revoke it. This will open a security alert and notify admins of this repository.

Cancel

Allow Secret

Code Scanning

- Busca **vulnerabilidades en el código fuente**, errores de seguridad, y malas prácticas de programación que puedan comprometer la aplicación.
- **¿Cómo funciona?**
- Usa herramientas de análisis estático (como **CodeQL**, SonarQube, etc.).
- Analiza el código en cada **push**, **pull request** o programación periódica.
- Identifica:
 - Inyecciones SQL
 - Cross-site scripting (XSS)
 - Desbordamientos de búfer
 - Accesos a archivos inseguros
 - Oportunidades de mejora en calidad del código

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Code scanning 2

Secret scanning

Code scanning

✓ All tools are working as expected

Tools 1 + Add tool

is:open branch:main

☐ 2 Open ✓ 0 Closed

Language Tool Branch Rule Severity Sort

☐

Reflected cross-site scripting

High

main

#2 opened 10 minutes ago • Detected by CodeQL in start.js :7

☐

Workflow does not contain permissions

Medium

main

#1 opened 10 minutes ago • Detected by CodeQL in .github/workflows/crear-discusion-commits.... :8

ProTip! You can configure CodeQL to run with additional queries. [Learn more](#)

Demo

- Dependencias: gráfico, alertas, actualizaciones de seguridad, actualizaciones de versiones

<https://github.com/skills/secure-repository-supply-chain>

Manejo de archivos confidenciales fuera del repositorio con .gitignore

- Es fácil para los desarrolladores pasar por alto archivos incluidos en un commit.
- A veces, estos archivos pasados por alto son benignos, como los archivos de compilación intermedios.
- Sin embargo, siempre existe el riesgo de que alguien pudiera comprometer accidentalmente datos confidenciales. Por ejemplo, alguien podría comprometer una clave de API o datos de configuración privados que un actor malintencionado podría usar.
- Una técnica que ayuda a evitar este riesgo consiste en compilar y mantener archivos .gitignore.
- Estos archivos indican a las herramientas cliente, como la utilidad de línea de comandos git, que omitan las rutas de acceso y los patrones al agregar archivos para una confirmación.

 [main](#) [Demos-Sesion-8](#) / [.gitignore](#)  icebeam7 Create .gitignore[Code](#) [Blame](#) 19 lines (15 loc) · 601 Bytes

```
1   # User-specific files - Ignore all files ending in ".suo"
2   *.suo
3
4   # Mono auto generated files - Ignore all files starting with "mono_crash."
5   mono_crash.*
6
7   # Build results - Ignore all files in these folders found at any folder depth
8   [Dd]ebug/
9   [Rr]elease/
10  x64/
11  x86/
12
13  # Root config folder - Ignore this directory at the root due to leading slash
14  # Removing the slash would ignore "config" directories at all depths
15  /config
16
17  # Ignore intermediate JS build files produced during TypeScript build at any
18  # folder depth under /Web/TypeScript. This won't ignore JS files elsewhere.
19  /Web/TypeScript/**/*.*.js
```

Reglas de protección de ramas

- Forzar workflows en una o más ramas a fin de requerir aprobación o pasar alguna validación de estado para un PR que desee integrarse en una rama.
- Use los flujos de trabajo que protegen la rama para:
 - Ejecución de una compilación para comprobar que se pueden compilar los cambios de código.
 - Ejecutar un linter para comprobar si hay errores tipográficos y se cumplen las convenciones de codificación internas
 - Ejecutar pruebas automatizadas para comprobar los cambios de comportamiento del código

Habilitar/Deshabilitar features en
tu cuenta de GitHub

Habilitar o deshabilitar funcionalidades en tu cuenta de GitHub

- GitHub permite activar o desactivar varias features de colaboración o seguridad, tanto a nivel de usuario como de organización/repositorio.

Algunas funciones que puedes habilitar/deshabilitar:

- Issues
- Discussions
- GitHub Pages
- Actions
- Wikis
- Dependabot alerts

¿Dónde se gestionan?

- Desde la pestaña **Settings** de cada repositorio o desde la configuración de tu cuenta/organización.

Q&A

¡Gracias!